DSA Coursework 2026

University Course Registration and Waiting List Management System

A university offers a course with limited seats.

- Students who successfully register for the course are stored in a **Linked List (Student List)**.
- When the course is full, additional students are placed in a **Waiting Queue** (FIFO).
- Students who drop the course are recorded in a **Drop History Stack** (LIFO) to support undo operations.

Your task is to implement all three data structures using **singly linked lists** and integrate them into a single system.

Requirements

1. Maintain student data

- Student ID
- Student Name

2. Student Registration List (Linked List)

Implement a linked list to:

- Add a student to the registered student list.
- Remove a student from the registered student list.
- Search for a student by Student ID.
- Display all registered students.

2. Waiting Queue (Linked List-Based Queue)

Implement a queue to:

- Add a student to the waiting queue when the course is full.
- Admit the next student from the waiting queue when a seat becomes available.
- Display all students in the waiting queue.

3. Drop History Stack (Linked List-Based Stack)

Implement a stack to:

- Push a student onto the stack whenever they drop the course.
- Pop the most recently dropped student to undo a drop.
- Display the drop history.

Rules of the System

- The course can accommodate **5 registered students**.
- If the registered list already contains 5 students, new students must be added to the **waiting queue**.
- When a registered student drops the course:
  - Remove the student from the registered list.
  - Push the student onto the drop history stack.
  - Automatically move the first student from the waiting queue to the registered list.
- If the administrator selects **Undo Last Drop:**
  - Pop the student from the drop history stack.
  - Add the student back to the registered list if space is available.

Sample Menu

Main Menu

1. Register Student

2. Drop Student

3. Undo Last Drop

4. Search Student

5. Display Registered Students

6. Display Waiting Queue

7. Display Drop History

8. Exit

Write a C program to fulfill this scenario. Submit on or before 1st of July, 2026.